When Does Even Spacing Beat a Trained Segmenter? Tolerance, Segment Length and Published Results

Anonymous ACL submission

Abstract

Segmentation and chaptering are scored by matching boundaries within a tolerance. We study one source of rank changes: the window relative to mean segment length. When this ratio is large, evenly spaced boundaries given the reference count can beat a trained segmenter; when it is small, every method scores lower on longer segments. At equal counts, a trained segmenter beat even spacing within one sentence but lost within three on lectures with short segments. Four preregistered tests, hash-locked internally, on over 15,000 unseen videos, mostly from explored label sources, and over 2,000 dialogues supported 26 of 33 claims: no directional claim failed, but the quantitative predictions on dialogue did. Chance-corrected WindowDiff favored the trained model in every length group in point estimates. Suppressing picks closer than a segment (post hoc) shrinks the reversal to 0.02–0.03 F1 where the window exceeds a segment. Under published localization metrics, even spacing with a count predicted from duration ties two Vid2Seq variants on VidChapters-7M at temporal IoU 0.5, though not Chapter-Llama; given the reference count it beats them at 0.5 but not at 0.7. Rankings at 3–5 second tolerances stand. We recommend reporting the ratio, an even-spacing baseline and a length-scaled metric.

1 Introduction

Automatic chaptering divides a long video into navigable parts. Many systems segment the transcript (Retkowski and Waibel, 2024) or work in the text domain with transcripts and frame captions (Ventura et al., 2025). Outputs are usually scored by matching predicted boundaries to reference boundaries within a tolerance, a number of sentences or seconds, or by window metrics such as Pk (Beeferman et al., 1999) and WindowDiff (Pevzner and Hearst, 2002).

A tolerance means little on its own; what matters is its width relative to the segments it is applied to. We write $r = w/L$ for the matching window w (for a collar of c sentences, $w = 2c + 1$ sentences; for $\pm t$ seconds, $w = 2t$) over the mean reference segment length L. When r is large, a baseline that knows only how many boundaries there are and spaces them evenly lands within the window of most reference boundaries, and a model that places boundaries well but sometimes bunches them can fall below it. When r is small, every method misses more, so scores fall as segments grow even if relative quality does not. A tolerance fixed in sentences or seconds lets r vary across documents, datasets and durations; a threshold relative to each segment, such as temporal IoU, fixes r, but possibly at a large value.

We met this problem while testing whether a segmenter's advantage transfers across datasets. On YTSeg, with the number of boundaries held equal, a reimplemented MiniSeg beat even spacing at every collar. On VT-SSum lectures, whose segments are shorter, a preregistered test found the advantage inconclusive at collar 1, and in exploratory follow-up it reversed at collars 2 and 3. Either the model does not transfer, or the metric changes meaning when segments get shorter. That fixed tolerances are lenient, and that evenly spaced baselines can be strong, is known (Section 2). What we have not found is a test of *when* the ranking flips, with the hypothesis fixed before the test data are seen, or a check of what this means for published results. Our contributions are:

1. **A ratio account of rank reversals, tested in four preregistered, sealed tests** (hash-locked internally, not publicly registered). Three on video, on unseen data from four datasets and two unread halves of a fifth, with five training seeds, two unsupervised segmenters and a count estimated without the reference, supported 22 of 23 directional claims, with one inconclusive (20 with Bonferroni-corrected

intervals); they replicate directions within label sources we had explored (except AVLectures, 15 courses). The fourth, on dialogue, supported its directional claims or left them inconclusive, and its quantitative predictions failed: 26 of 33 claims supported, 4 inconclusive, 3 falsified (Sections 4, 5 and 8).

2. **A coverage reading** of the account, based on a standard renewal argument: even spacing wins by spreading its boundaries, not by aligning with the reference, and with post hoc suppression of picks closer than a segment the reversal shrinks to 0.02–0.03 F1 and remains only where the window exceeds a segment. On dialogue the predicted sign appeared, by about 0.01 F1, but the model's predicted sizes and flip points did not (Sections 4, 7 and 8).

3. **Published results rescored**: where the ratio is large, even spacing given the reference count matches or beats published systems under their own localization metrics at loose thresholds (with a count from duration, it ties some); where it is small, published rankings stand, but a published decline on long videos is matched by a predictor of constant relative quality; in 76 published tables, 15 claims flip with the tolerance, and the ratio is one of several sources of such flips (Section 6).

Section 5 reports the three preregistered tests on video and Section 8 the fourth, on dialogue; Table 2 lists the claims of all four. An AI agent wrote the four preregistrations; an author read those of Tests 1 to 3 before scoring, and the agent approved Test 4 under authority the authors delegated (Appendix B). We count the four tests alike, because each was hash-locked before scoring. Section 4 produced the hypotheses of Tests 1 to 3 on open data. Fits of the coverage model, Section 6, the checks of Section 7 and the decision replay are post hoc or exploratory, and we label them so.

2 Related Work

Segmentation metrics. Pk and WindowDiff slide a window of half the mean reference segment length, so their tolerance scales with length. Their biases are well studied (Pevzner and Hearst, 2002; Georgescul et al., 2006; Lamprier et al., 2007); Niekrasz and Moore (2010) showed that they favor segmentations with few or adjacent boundaries and proposed k-κ, an unbiased measure that corrects for chance agreement, the kind of normalization we use. Other measures include WinPR (Scaiano and Inkpen, 2012), a multi-annotator WindowDiff (Kazantseva and Szpakowicz, 2012), segmentation similarity (Fournier and Inkpen, 2012) and boundary similarity (Fournier, 2013), which YTSeg reports beside F1 and Pk. In music structure analysis, listeners' ratings of boundaries did not follow the tolerance F-measure (Nieto et al., 2014), whose standard implementation is mir_eval (Raffel et al., 2014); metric validity is a broader concern (Mathur et al., 2020).

Tolerances and simple baselines. That wide tolerances favor uniform baselines is not new. Closest to our work, Coen (2025) shows for dialogue topic segmentation that window-tolerant F1 is tightly coupled to boundary density, and compares methods with periodic baselines that know the reference count. We differ in three ways: we fix the ratio of tolerance to mean segment length as the variable and predict the sign of the contrast within datasets before seeing test data; we test whether the reversal survives decoding changes and estimated counts; and we rescore published video results, where the question is which reported claims depend on the ratio. Benchmark artifacts of a similar kind are known in video: moment retrieval benchmarks reward predictions that follow dataset priors (Otani et al., 2020), and the standard dense captioning scorer (Krishna et al., 2017) rewards hundreds of redundant captions (Fujita et al., 2020); benchmark conclusions can hinge on such choices (Dehghani et al., 2021). Equal splitting into the reference number of segments is a strong baseline for lectures (Singh S et al., 2023) and competitive in unsupervised action segmentation, where Sarfraz et al. (2021) introduced it as Equal Split. Chaptering uses tolerances in seconds and temporal IoU thresholds (Yang et al., 2023a; Ventura et al., 2025); generic event boundary detection uses a tolerance relative to the video's duration (Shou et al., 2021), while our local relative F1 scales the tolerance with the neighbouring segments.

Chaptering. YTSeg and MiniSeg (Retkowski and Waibel, 2024) provide a benchmark of spoken YouTube transcripts and an efficient hierarchical segmenter. VidChapters-7M (Yang et al., 2023a) provides creator chapters at scale; VT-SSum (Lv et al., 2021) and AVLectures (Singh S et al., 2023) cover lectures, and AMI (Carletta et al., 2006) and ICSI (Janin et al., 2003) cover meetings, with topic

Set	Labels	Docs	Unit	Seg.	CV
Open, exploratory					
YTSeg test	creator chapters	1,447	ch.	21.8	.49
VT-SSum test	slide groups	958	video	8.8	.81
Sealed, Test 1					
AVLectures	contents, clips	348	course	89.5	.38
VidChapters-7M	creator chapters	7,357	ch.	29.2	.58
AMI, ICSI	topic levels	25–136	series	65–247	.8–1.1
Sealed, Tests 2 and 3					
VT-SSum train, half A	slide groups	3,682	video	9.0	.81
VT-SSum train, half B	slide groups	3,672	video	8.9	.81

Table 1: Data. Docs: documents scored. Unit: the cluster resampled in the bootstrap (ch.: channel). Seg.: mean reference segment length in sentences. CV: median over documents of the coefficient of variation of segment length (computed post hoc). Test 4's dialogue sets are in Appendix H. Meetings were secondary (Appendix C).

segmentations distributed with AMI and, for ICSI, by Galley et al. (2003). TextTiling (Hearst, 1997) and C99 (Choi, 2000) are classic unsupervised segmenters; Koshorek et al. (2018) cast text segmentation as supervised learning. Retkowski et al. (2026) score audio chaptering with F1 at a 6-second resolution and report that performance degrades sharply for all models beyond 20 to 30 minutes. Preregistration (Nosek et al., 2018) has been proposed for NLP by van Miltenburg et al. (2021).

3 Setup

Data. Table 1 lists the sets. Subtitles and Whisper transcripts were split into sentences by one rule-based splitter (precision 0.99 and recall 0.95 against YTSeg's sentences); meetings use dialogue acts. A boundary goes to the sentence whose start is nearest the labelled time. Documents with fewer than two sentences or no internal boundary drop out (569 of 7,926 VidChapters videos); two of AVLectures' 350 segmented lectures were dropped because their subtitle and segment timelines disagreed. Appendix B explains every document count used in the paper. VT-SSum train was split by a hash of the file name into two halves; each was read only by the frozen code of one test.

Methods. We reimplemented MiniSeg (Retkowski and Waibel, 2024): a fine-tuned MiniLM sentence encoder (Wang et al., 2020; Reimers and Gurevych, 2019), a 12-layer document transformer with a causal mask, and a per-sentence boundary head (Appendix A). It is trained on YTSeg only, with five seeds (YTSeg test F1 44.3 to 44.8 in online mode; the original's YTSeg-only offline model reports 43.4). Every method places the reference count K in each document, so only placement differs: **MiniSeg** picks the K sentences with the largest boundary margin; **even** spacing places K boundaries evenly by sentence index (or by time, where stated); **random** places them at seeded random positions. Test 3 adds TextTiling, C99 and a count not taken from the reference.

Metrics. **Collar c F1**: micro F1 with one-to-one matching within c sentences: predicted boundaries, in order, each take the nearest unused reference (maximum matching changed no sign; Appendix C). **Chance-corrected WindowDiff (WD)**: $(s - s_r)/(1 - s_r)$, with $s = 1 -$ WindowDiff and s_r the score of random placement on the same documents. **Local relative F1**: a reference boundary may be matched within 0.2 times the shorter of its two neighbouring segments, chance corrected. **Boundary similarity B** (Fournier, 2013) with near-miss window n_t. Every contrast is a method minus even spacing on the same documents. Length groups use each document's mean segment length (≤ 6, 6–10, 10–16, 16–25, > 25 sentences); "short" means ≤ 10 and "long" > 16. Intervals come from a cluster bootstrap (1,000 resamples, unit in Table 1).

Preregistration. Before each sealed run, the preregistration, analysis code, prepared data and checkpoints were pinned by SHA-256, and the code refused to run if a hash differed. These locks are internal: file times put each freeze before scoring, but no external timestamp proves it (Appendix B). Each training seed had its own bootstrap. A claim that a quantity exceeds a threshold is **supported** if the lowest lower bound of the 95% intervals across seeds exceeds it, **falsified** if the highest upper bound falls below it, and **inconclusive** otherwise; "below" claims mirror this. We also report Bonferroni-corrected intervals. Each set was scored once.

4 The Account on Open Data

This section uses the opened YTSeg and VT-SSum test splits and is exploratory; it produced the hypotheses of Section 5. On YTSeg, MiniSeg beat even spacing by 0.295 F1 at collar 1 and 0.119 at collar 3. On VT-SSum the advantage shrank to 0.056 at collar 1 and reversed to -0.028 and -0.079 at collars 2 and 3. Grouping videos by their mean segment length showed the same rever-

sal inside each dataset, YTSeg included (Figure 1; Table 6 in Appendix C), so within these data the ratio, not the dataset, sets the sign. Removing a random share of VT-SSum's reference boundaries until segments averaged 21 sentences, close to YTSeg, moved the collar-3 contrast from -0.079 to -0.011.

Why even spacing wins: coverage. With K known, one might expect the j-th evenly spaced boundary to stay near the j-th reference boundary. It does not: the gap grows with the accumulated variation in segment length and, at the middle boundary, has a median standard deviation of 9 sentences on YTSeg and 17 on VT-SSum. Even spacing scores through *coverage* instead. Its boundaries are spread one per L sentences, and each is matched whenever its window of w sentences holds any reference boundary. Treating segment lengths G as a renewal process gives the standard coverage result

$$F_1^{\text{even}}(c) \approx 1 - \frac{\mathbb{E}\left[(G-w)_+\right]}{\mathbb{E}[G]}, \qquad w = 2c+1, \tag{1}$$

which, if the even grid's phase is random relative to the references, depends only on $r = w/L$ and the spread of G, and at $r = 1$ is near 1 when segment lengths barely vary but about 0.63 when they are exponential (CV $= 1$); at collar 3, $w = 7$ exceeds a whole segment in the shortest groups. A trained model gains from a wider window only through the boundaries it misplaces, so the ranking flips where the two curves cross. Chance correction cannot prevent this, because it rescales scores within a group but keeps the order of methods. With few boundaries and regular segments the phase is not random and it under-predicts (Appendix F). We use Equation 1 as a tool, not as a contribution; we formalized it after the sealed tests, and the preregistered claims rest on the segment-length account alone.

Choosing a length-scaled metric. Before looking at MiniSeg's scores, we compared metrics on a hypothetical predictor that places half its boundaries correctly and the rest at random, whose quality therefore does not depend on length. A rule fixed on development splits chose the metric whose score for this predictor varied least across length groups. On the test splits, chance-corrected WD varied by 0.033, local relative F1 by 0.046, Pk by 0.058 and collar 3 by 0.258. Local relative F1 kept the order of five methods most stable across groups (mean Kendall tau 0.920, against 0.570 for collar 3; Appendix C). WD kept even spacing near chance in every group (at most 0.075), so it cannot produce the reversal itself, and it penalizes bunched boundaries rather than rewarding them (Appendix C). This choice assumes that an error matters in proportion to segment length; Section 8 returns to this assumption.

5 Preregistered Tests on Video

Test 1: four unseen datasets. The primary sets were AVLectures, whose labels come from lecture contents, and VidChapters-7M, which shares YTSeg's label source but has new videos and Whisper transcripts; AMI and ICSI meetings were secondary. Nine claims covered transfer (P1–P3), the length account (P4) and rank stability (P5). One deviation occurred: the first run stopped with an out-of-memory error before writing any result, and the fix, which left all scores unchanged on open data, was approved and disclosed (Appendix B).

Test 2: unread slide-labelled videos. AVLectures segments are almost all long, so Test 1 could not check the reversal on a second label definition. Test 2 used half of the VT-SSum train split, which no code had read, with six claims, two of them on the width of boundary similarity's window: widening B's near-miss window from two to five sentences raises r and should bring the reversal back.

Test 3: other segmenters and an estimated count. Two objections remained: all results came from one model, and every method was given K, which the even baseline needs. Test 3, on the other VT-SSum train half, added TextTiling and C99 with textbook settings, each placing its top K positions, and a second setting in which every method placed $\hat{K}$, the number of sentences with a positive MiniSeg margin. It was requested after Test 2's results were seen, and its claims were set from an exploratory run on half A.

Results. Table 2 lists all claims. Of the 23 on video, twenty-two were supported and one was inconclusive: AVLectures' WindowDiff advantage (every seed between 0.105 and 0.121) had a lowest interval bound of 0.009 against a threshold of 0.05, with only 15 courses. Because a bootstrap over 15 clusters is coarse, we also counted courses post hoc: MiniSeg led even spacing at collars 1 and

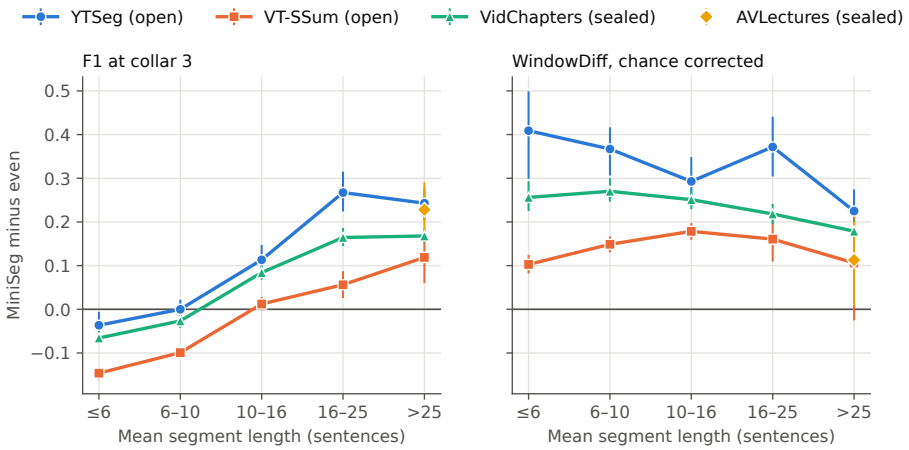


Figure 1: MiniSeg minus even spacing by mean segment length. Sealed sets: medians over five seeds, bars span the 95% intervals across seeds; open sets: seed 0. Collar 3 reverses in short segments in every set with short videos, while chance-corrected WindowDiff favors MiniSeg throughout. AVLectures has one point because 344 of its 348 lectures fall in the longest group.

3 in 14 of 15 courses (one tie; one-sided exact sign test, tie counted as a loss, $p = 0.0005$) but under WD in only 9 ($p = 0.30$), in line with the inconclusive verdict. By the preregistered reading, transfer was partial and the length account was supported. All Test 2 and Test 3 claims also held with Bonferroni-corrected intervals; in Test 1, P1 and P2 on AVLectures and P5 did not, leaving 20 of 23 supported. With Test 4, 26 of 33 claims were supported, 4 were inconclusive and 3 were falsified, all three quantitative predictions on dialogue.

The sealed data repeat the open pattern (Figure 1). On 7,357 VidChapters videos, collar 3 put MiniSeg 0.066 below even spacing when segments averaged at most six sentences and 0.168 above it beyond 25, while WD favored it by 0.18 to 0.27 in every group. On the unread VT-SSum videos of Test 2, the collar-3 contrast ran from -0.137 to $+0.138$ across length groups, while WD stayed between $+0.107$ and $+0.187$, and widening B's window to five sentences brought the reversal back (V5, V6). Meetings, whose segments are all long (65 to 247 dialogue acts), favored MiniSeg at every level and metric in point estimates; they fit the account but cannot test it. In Test 3, C99 showed the MiniSeg pattern at a smaller scale (collar 3 from -0.033 to $+0.072$, WD between $+0.057$ and $+0.065$; Appendix C), and TextTiling, below even spacing in every group, still closed its gap with length faster under collar 3 than under WD (T6). With $\hat{K}$, far below K (median ratio 0.08), segments were much longer than the collar and MiniSeg led in every group, as T8 predicted.

6 Published Results Through the Ratio

The account makes two predictions for published work. Where r is large, a trivial spreading baseline should match or beat published systems; where r is small, published rankings should stand, but scores should fall with segment length for every method. We checked both post hoc, without author outputs except where authors had released them (Appendix G), and only for localization, since even spacing writes no titles. Even spacing used either the reference count, an oracle the published systems lack, or a count predicted from duration alone by a fit on training videos; we lead with the latter.

Large ratio: a trivial baseline catches up. Scored by temporal IoU at 0.5 ($r \approx 0.67$), as in Table 4 of Yang et al. (2023a), even spacing with a count from duration alone tied the 43.2 reported there for two fine-tuned Vid2Seq (Yang et al., 2023b) variants and beat PDVC (Wang et al., 2021) (37.6); with the reference count it beat every system in that table (55.5 against 45.5), but not at tIoU 0.7, where four Vid2Seq variants led (up to 27.4 against 24.5; Table 3). Chapter-Llama reports scores above it at both thresholds. Most of the gain came from spreading, on regular chapters (Appendix G). Under the protocol of Chapter-Llama (Ventura et al., 2025), whose main claim stands (45.3 F1 against 24.9 for even spacing with K), the fine-tuned Vid2Seq it compares against fell below even spacing with K on medium and long videos. On AVLectures, rescoring the outputs that Singh S et al. (2023) released, their method's lead

ID	Arm, set	Quantity	Est.	95%	
Test 1: AVLectures (A), VidChapters (V), MiniSeg					
P1	A	WD $> .05$	.115	.009, .204	I
P1	V	WD $> .05$	.227	.213, .239	S
P2	A	local $> .05$	.185	.077, .273	S
P2	V	local $> .05$	.211	.200, .223	S
P3	A	C1 $> .05$	.198	.112, .297	S
P3	V	C1 $> .05$	.192	.183, .201	S
P4a	A	C3 > 0	.228	.159, .290	S
P4b	V	C3, L$-$S > 0	.213	.197, .229	S
P5	A+V	τ WD$-$C3 > 0	.160	.053, .293	S
Test 2: VT-SSum train half A, MiniSeg					
V1	short	C3 < 0	$-.117$	$-.124, -.109$	S
V2	all	C3, L$-$S > 0	.214	.195, .238	S
V3	≤ 6	WD > 0	.107	.094, .121	S
V4	all	WD $> .05$	.145	.138, .155	S
V5	≤ 6	$B_5 < 0$	$-.149$	$-.165, -.133$	S
V6	≤ 6	$B_2-B_5 > 0$	.156	.147, .165	S
Test 3: VT-SSum train half B					
T1	C99	C3, L$-$S > 0	.102	.090, .114	S
T2	C99, short	C3 < 0	$-.025$	$-.028, -.023$	S
T3	C99, ≤ 6	WD > 0	.057	.048, .066	S
T4	TextTiling	C3, L$-$S > 0	.116	.103, .128	S
T5	C99	(C3$-$WD), L$-$S > 0	.102	.087, .117	S
T6	TextTiling	(C3$-$WD), L$-$S > 0	.097	.077, .116	S
T7	MiniSeg, $\hat{K}$	C3, L$-$S > 0	.041	.030, .053	S
T8	MiniSeg, $\hat{K}$	C3, short > 0	.019	.015, .023	S
Test 4: SuperDialseg; MiniSeg (M) and cross-encoder (X), each judged					
H1	M, X; $L \leq 2.5$	C2, C3 < 0	$\leq -.009$	$-.021, -.001$	S, S
H2	M, X; all	WD > 0	$\geq .51$	$\geq .35$	S, S
H3	M, X	corr. $> .8$, sign $\geq .8$	.74, .76		F, F
H4	M, X	signs as predicted	11/14, 10/12		I, I
H5a	even	Eq. 1 MAE $< .03$	.077	.067, .094	F
H5b	even	simulation MAE $< .03$	.022	.019, .033	I

Table 2: All preregistered claims. Estimates are arm minus even spacing, medians over five runs (three in Test 4); intervals span the lowest lower and highest upper 95% bound across runs. Test 4 (Section 8, Table 15): H3 shows each model's share of cells with the predicted sign, H4 the cells signed as predicted (none falsified), H5 the mean absolute error of predicted even-spacing F1. C1, C3: collar F1; local: local relative F1; B_n: boundary similarity at $n_t = n$; L$-$S: long (over 16 sentences) minus short (up to 10); τ: mean Kendall tau of five methods' order across dataset and length groups. S: supported, I: inconclusive, F: falsified.

Benchmark (system)	Metric	r	Pub.	No lab.	Even K
Large ratio ($r \geq .67$)					
VidChapters (best Vid2Seq)	F1@tIoU .5	.67	45.5	43.2	55.5
AVLectures (theirs)	BS@300 s	1.1	84.8	78.1	86.1
Intermediate or mixed ratio					
VidChapters (best Vid2Seq)	F1@tIoU .7	.35	27.4	16.1	24.5
VidChapters 30–60 min (Vid2Seq)	F1 (C-L)	.03–.67	16.7	14.6	23.6
Small ratio ($r \leq .11$)					
VidChapters (best Vid2Seq)	F1@5 s	.06	33.1	18.8	20.2
AVLectures (theirs)	BS@30 s	.11	58.7	30.1	32.5
YTSeg (MiniSeg)	F1	.05	43.4	8.2	7.8

Table 3: Published systems (Pub.) and even spacing under each benchmark's published evaluation (post hoc). r: window over the set's mean segment length. For a tIoU threshold θ, a predicted segment of the right length passes when its start is within $L(1-\theta)/(1+\theta)$, so $r = 2(1-\theta)/(1+\theta)$; this is an upper bound, since a prediction of the wrong length must lie closer (C-L averages θ from 0.5 to 0.95, as in action detection mAP; Alwassel et al., 2018). F1@tIoU .5, .7: harmonic mean of the precision and recall in Table 4 of Yang et al. (2023a). C-L: Chapter-Llama scorer (Ventura et al., 2025). AVLectures: the authors' released outputs; BS@t: boundary score within $\pm t$ seconds. No lab.: count from duration; Even K: reference count (oracle).

Predictor	F1 kept	Pk change
Constant quality, s=.05–.25	.22–.34	-8 to -1%
$s = .10$	.28 (.21, .34)	-1%
Published (3 systems)	.27–.34	$+30$ to $+36\%$
MiniSeg, ours	.48 (.38, .57)	$+29\%$
Random times	.11	$+6\%$

Table 4: Videos over 60 minutes against videos under 10 minutes, `chunkseg` with 6-second chunks. F1 kept: F1 at ≥ 60 min divided by F1 at < 10 min. Pk change: relative change (lower is better). Intervals: cluster bootstrap over channels.

over equal splits given the reference count shrank as the window widened, from 26.1 points at ± 30 seconds ($r \approx 0.11$) and 14.3 at ± 120 to a tie at ± 300 (-1.3, -3.8 to 1.1; $r \approx 1.1$). Per course, it led in all 15 at ± 30 and ± 120 seconds and in 7 at ± 300. At ± 120 seconds the lead grew with mean segment length, from $+6.8$ for segments under six minutes to $+19.1$ for twelve minutes or more, as Equation 1 predicts.

Small ratio: rankings stand, durations do not compare. At the usual chaptering tolerances, r is small and published rankings stand: on YTSeg (Retkowski and Waibel, 2024; Retkowski et al., 2026), and for every fine-tuned VidChapters system at 3 and 5 seconds, even spacing scored at the level of random placement (VidChapters F1@5 s 20.2 against 19.4). A small ratio has its own failure: as videos grow, chapters grow and r shrinks further. Retkowski et al. (2026) report that chaptering quality falls sharply beyond 20 to 30 minutes, scored by F1 over 6-second chunks with their scorer `chunkseg`. With their scorer, bins and settings on YTSeg test (1,403 videos), we scored a predictor that moves each reference boundary by a normal error with standard deviation a fixed share s of the shorter neighbouring segment, so its relative quality never changes. From under 10 minutes to over an hour, it kept 22% to 34% of its F1 for s from 0.05 to 0.25, and the three published systems kept 27% to 34% (Table 4). F1 at 6 seconds therefore cannot separate a weaker model from a stricter metric. Pk, whose window scales with segment length, stayed flat for the simulated predictor but worsened by 30% to 36% for the published systems, so real systems do degrade, although a 30% rise in Pk is not a 30% loss of quality.

Published tables. We transcribed 76 tables from 30 papers that report the same systems under several tolerances or metrics. In the 45 tables with several tolerances, 14% of system pairs swapped order between the tightest and widest (10% when both gaps were at least half a point), and in 15 cases from 12 papers a claim in the text held at one reported setting and failed at another. For example, Singh S et al. (2023) write that equal splits outperform K-Means and CTE, which holds on segment-overlap F1 (74.0 against 55.7 and 57.3) and reverses at ±30 seconds (32.5 against 44.9 and 41.5). Chaptering tolerances of 3 to 5 seconds, a few percent of a chapter, flipped only 2.8% of pairs and never the winner, as Equation 1 predicts. Across all tables, flips rose with r (Spearman $\rho = 0.32$) but more with the distance between the two settings ($\rho = 0.66$), and temporal IoU thresholds flipped most often (29%), so the ratio is one source of rank changes, not the only one.

7 How General Is the Account?

These checks are post hoc or exploratory; details are in the appendices.

Decoding. MiniSeg takes the K largest margins, which can bunch: 30.7% of its consecutive picks on VT-SSum were adjacent, against 14.7% of reference boundaries. Skipping picks within g sentences of an earlier pick, a fixed-radius non-maximum suppression ($g = 1$ to 5, all post hoc), shrank the reversal and moved it to shorter segments. At 6 to 10 sentences per segment it vanished for $g \geq 2$, and collar 3 favored MiniSeg on every whole set. In the shortest group (at most six sentences, $r > 1$), every radius left MiniSeg below even spacing on VT-SSum test, both sealed halves and VidChapters (every interval excluding 0; on YTSeg only for $g \geq 4$; Table 11), while WD favored it throughout. The deficit was 0.022 to 0.033 at $g = 2$ and 3 and up to 0.079 at $g = 4$ and 5, radii that can suppress true neighbouring boundaries in segments of at most six sentences. The selection rule thus sets how far the reversal reaches; it does not remove it once the window exceeds a segment. Tolerances in words, which remove the sentence splitter, gave the same picture (Appendix C).

Counts. The reversal needs a count near the reference, not the reference itself. With a MiniSeg threshold fit on VT-SSum dev so that predicted and reference totals match (about $1.17K$), collar 3 put MiniSeg 0.078 below even spacing for segments up to six sentences and 0.065 above beyond 25, with WD positive in every group (Appendix C).

Set, arm	Metric	≤ 6	6–10	> 25	L−S
VT-SSum, LLM 4B	C3	$-.175^{*}$	$-.074^{*}$	$+.129^{*}$	$+.200^{*}$
	WD	$+.087^{*}$	$+.174^{*}$	$+.126^{*}$	$-.015$
VT-SSum, LLM 14B	C3	$-.074^{*}$	$-.013$	$+.186^{*}$	$+.205^{*}$
	WD	$+.218^{*}$	$+.248^{*}$	$+.153^{*}$	$-.061$
VT-SSum, MiniSeg	C3	$-.130^{*}$	$-.086^{*}$	$+.115^{*}$	$+.192^{*}$
	WD	$+.164^{*}$	$+.178^{*}$	$+.089$	$-.047$
YTSeg, LLM 4B	C3	$-.089^{*}$	$-.099^{*}$	$+.198^{*}$	$+.260^{*}$
	WD	$+.310^{*}$	$+.296^{*}$	$+.238^{*}$	$-.074$
YTSeg, LLM 14B	C3	$-.056^{*}$	$-.014$	$+.161^{*}$	$+.234^{*}$
	WD	$+.342^{*}$	$+.464^{*}$	$+.161^{*}$	$-.187^{*}$
YTSeg, MiniSeg	C3	$-.019$	$-.032$	$+.230^{*}$	$+.269^{*}$
	WD	$+.403^{*}$	$+.419^{*}$	$+.233^{*}$	$-.067$

Table 5: Zero-shot language models (LLM; Qwen3 4B and 14B) and MiniSeg minus even spacing at the reference count, on the same 150 videos per set, by mean segment length (exploratory, one bootstrap). *: 95% interval excludes 0.

Other models and annotators. Zero-shot Qwen3 models (Qwen Team, 2025) at 4B and 14B parameters, prompted to list topic starts in windows of 60 numbered sentences and keeping the K most often listed, showed the same reversal as MiniSeg on 150 opened videos each from VT-SSum and YTSeg (Table 5): below even spacing at collar 3 in the shortest group, above it beyond 25 sentences, and above it under WD in every group. The 14B model's deficit was about half the 4B model's.

Human topic segmentations of ICSI meetings scored against each other showed the same narrowing over even spacing as the window grew, from 0.37–0.59 at $c/L = 0.05$ to 0.12–0.30 at 0.5. At the human's own count the gap never closed, but against even spacing given the reference count it turned negative at $c/L = 1$ in three of four annotator pairings, significantly in two (Appendix F).

8 Discussion

Predicting when the ranking flips. Post hoc, Equation 1 predicted even-spacing F1 on every dataset and length group with a mean absolute error of 0.017 from L and the coefficient of variation of segment length alone (correlation 0.996), while the alignment view missed by 0.20. Modelling MiniSeg as placing a share p of its boundaries within one sentence, fitted at collar 1, and the rest by chance, $F_1^{\text{model}}(c) \approx p + (1-p)\,[1 - e^{-(2c+1)(1-p)/L}]$, predicted MiniSeg minus even at

collars 2 and 3 with correlation 0.96 over 52 cells, the right sign in 43 of 46 cells above 0.02 in size, and the collar-3 reversal at 9 to 10 sentences per segment against an observed 8 to 13 (Appendix F). It explains when the ranking flips better than by how much: in the shortest VT-SSum segments it predicted -0.01 to -0.02 where -0.13 to -0.15 was observed, and on dialogue (below) it erred the other way.

Test 4: a prospective test on dialogue. To test both models prospectively, we preregistered predictions for topic segmentation of written dialogue, where segments last 2 to 8 utterances: SuperDialseg (Jiang et al., 2023), TIAGE (Xie et al., 2021) and DialSeg711 (Xu et al., 2021). Each model's share p was fixed on development splits, and test contrasts were predicted from each test group's segment lengths (Appendix H). Two segmenters trained on SuperDialseg, MiniSeg and a RoBERTa cross-encoder (Liu et al., 2019) with three seeds each, beat even spacing by 0.47 to 0.74 F1 at collar 0 but fell 0.009 to 0.013 below it at collars 2 and 3 in the shortest dialogues ($r \geq 2$), while WD favored them in every group; both claims were supported for both models (Table 2). Where a contrast was predicted at least 0.02 in size, its sign was right or inconclusive, never falsified (H4, inconclusive for both). The quantitative claims fared worse. Predicted and observed contrasts correlated at 0.83 to 0.84, but signs agreed in only 74 to 76% of cells against a preregistered 80% (falsified). The flip came at $r \approx 1.7$ to 2.3 instead of 1.1, because the trained models again gained more from wider windows than the mixture allows. Once $w \geq L$, Equation 1 in closed form over-predicted even spacing by 0.077 (falsified, as development data had shown), while an exact simulation from the same segment lengths erred by 0.022 (inconclusive against 0.03). The ratio predicts the direction of the flip; its location and size need a better model of the trained segmenter, and the quantitative predictions failed, two of them as dev checks recorded in the preregistration had suggested (H3 for MiniSeg, H5a). Unlike Tests 1 to 3, this preregistration was approved by the agent under delegated authority and read by the authors after scoring (Appendix B).

When the criteria disagree. We replayed decisions on already scored data (post hoc; Appendix I). The costs below are measured by other metrics, so they show how far the criteria disagree, not which one is right. Choosing between MiniSeg and even spacing on VT-SSum dev by collar-3 F1 picked even spacing for all five seeds; WD, local relative F1 and collar 1 picked MiniSeg. On VT-SSum test and both sealed halves, the collar-3 choice lost 0.144 to 0.150 WD (every interval excluded 0); without the reference count, 0.153 to 0.156. With the reference count, the choices differed only up to 16 sentences per segment ($r \geq 0.57$), never on whole YTSeg, VidChapters or AVLectures; count error is a second route (Appendix I). With peak-suppressed MiniSeg among the candidates the cost fell to 0.006: it needs a spreading baseline among them.

Which tolerance is right? Choosing a length-scaled metric assumes that an error matters in proportion to segment length; a viewer who jumps to a chapter may care about absolute seconds instead. Under either view, what a fixed tolerance measures depends on the ratio: in seconds, the reversal appeared once the window neared the chapter length (30 seconds for chapters up to two minutes) and not at 3 or 5 seconds (Appendix C). We have no user or downstream study to say which view fits chaptering.

Recommendations. As reporting checklists do (Dodge et al., 2019): (1) Report the ratio of the matching window to segment length, the distribution of segment length, and the F1 of even spacing, which follows from segment lengths alone (by Equation 1, or by simulation once $w \geq L$); a score near it says that placement adds little. Subtracting it does not give a better ranking metric, because it is the same for every method on a set (Appendix J). (2) Report a length-scaled metric, such as chance-corrected WindowDiff or a relative-tolerance F1, beside any fixed tolerance, and do not compare fixed-tolerance scores across durations. (3) Include even spacing, with the reference count and with a count from duration; it is cheap and shows how much of a score the window gives away. (4) Compare placement at an equal count and report count error separately. (5) State the boundary selection rule: top-K with and without peak suppression gave different conclusions at medium lengths.

9 Conclusion

Which segmenter wins depends in part on the matching window relative to segment length: when it is large, a baseline that only spreads the right

number of boundaries can beat a trained model. Four preregistered tests replicated its direction, but the quantitative predictions on dialogue failed; suppressing nearby picks narrows the reversal to where the window exceeds a segment. Duration-only even spacing ties two Vid2Seq variants as reported by Yang et al. (2023a) at temporal IoU 0.5; rankings at 3 to 5 seconds stand.

Limitations

Replication sources. VidChapters shares its label source with YTSeg, and Tests 2 and 3 share theirs with the opened VT-SSum test, so each replication is on unseen videos within a source already studied. AVLectures has a different label definition but only 15 courses and long segments. **Models.** The preregistered tests use one trained architecture, trained only on YTSeg, and two untuned unsupervised segmenters; the language model checks use one model family up to 14B parameters, one prompt chosen on opened videos, and 150 videos per set. **Counts.** The preregistered comparisons give methods K or $\hat{K}$; calibrated counts were checked only post hoc. In Test 3, $\hat{K}$ was about $0.08K$, so segments far exceeded the collar and claim T8 was nearly certain to hold. **Decoding.** Peak suppression was checked post hoc, with the radius range chosen after seeing the results. **Published results.** Section 6 is post hoc. Even spacing with the reference count has an oracle advantage, which the count from duration removes. Mean temporal IoU has a high floor (random placement with K scored 52 under the Chapter-Llama scorer). We rebuilt the VidChapters ground truth without the authors' tokenizer filter. The tables were chosen because they report several settings, so their flip rates are not population estimates. For the long-video analysis we found no public outputs, so the published systems were not rescored; their split is not stated, YTSeg sentence times are interpolated, and the simulated error model is an assumption. **Units and language.** Collars count sentences, so results depend on the sentence splitter; we used one splitter and did not vary it, and splitting differs between subtitles, Whisper output and dialogue acts. All data come from English-language sources. **Beyond the ratio.** At equal segment length MiniSeg's advantage still differed between datasets, so the ratio does not explain all differences (Appendix F). **Statistics.** We did no power analysis before Test 1, whose AVLectures part has 15 courses. VT-SSum intervals resample videos; talks from one series may be correlated (Appendix C). **Timing.** Boundary times come from each dataset's own subtitles or Whisper output and are snapped to sentence starts; we had no independently timed reference. Poorly split VidChapters videos concentrate in the shortest sentence group, although the reversal held without them (Appendix C). **Preregistration.** The hash locks are internal and not publicly registered. An author read the preregistrations of Tests 1 to 3 before scoring; Test 4's was approved by the agent and read by the authors after scoring. **Research path.** The project began as a study of when to rerun a causal segmenter under a call budget (Appendix K); the ratio account grew out of its negative results on open data. **Dialogue test.** SuperDialseg boundaries fall almost only before user turns, which both models learn, so their share p is high and dialogue does not test a weak in-domain model (Appendix H); the models are near chance on TIAGE, and each has three seeds. **Decision replay.** It reuses already scored sets, and only the best and last checkpoints of each run were saved, so a full checkpoint trajectory was not replayed. **What counts as right.** We choose length-scaled metrics by an assumption, not by user or downstream evidence; users of chapters may care about absolute time errors. **Exposure.** All hypotheses come from data we had opened; the sealed tests confirm them on new data but were designed with knowledge of the open results. Test 3 was requested after Test 2's results were seen, and its claims come from an exploratory run on half A.

Ethics Statement

This work recruited no human participants or annotators. It reuses transcripts and annotations that their creators released publicly under the licenses in Appendix A, and involves no interaction with people and no private data, so we did not seek ethics board review.

Use of AI Assistants

An AI coding agent, of the kind studied as automated research systems (Lu et al., 2024), wrote the analysis code, wrote all four preregistrations, ran the experiments and drafted the text under the authors' direction. Before scoring, an author read the preregistrations of Tests 1 to 3 and asked for Test 1's to be frozen unchanged; for Test 4 the authors delegated approval to the agent, which

recorded its approval before scoring, and read the preregistration after scoring, finding no problem (Appendix B). All code and the manuscript were reviewed by the authors before submission, and the authors take full responsibility for the methods, results and writing. Errors found during the work, some of which would have changed conclusions, are listed in Appendix K. The preregistrations, freeze lists, frozen code and sealed results are in the supplementary material, with a script that checks their hashes; all code will be released.

References

Humam Alwassel, Fabian Caba Heilbron, Victor Escorcia, and Bernard Ghanem. 2018. Diagnosing error in temporal action detectors. In *Proceedings of ECCV*, pages 264–280.

Doug Beeferman, Adam Berger, and John Lafferty. 1999. Statistical models for text segmentation. *Machine Learning*, 34(1–3):177–210.

Jean Carletta, Simone Ashby, Sebastien Bourban, Mike Flynn, Mael Guillemot, Thomas Hain, Jaroslav Kadlec, Vasilis Karaiskos, Wessel Kraaij, Melissa Kronenthal, Guillaume Lathoud, Mike Lincoln, Agnes Lisowska, Iain McCowan, Wilfried Post, Dennis Reidsma, and Pierre Wellner. 2006. The AMI meeting corpus: A pre-announcement. In *Machine Learning for Multimodal Interaction (MLMI 2005)*, volume 3869 of *LNCS*, pages 28–39.

Freddy Y. Y. Choi. 2000. Advances in domain independent linear text segmentation. In *Proceedings of NAACL*, pages 26–33.

Michael H. Coen. 2025. When F1 fails: Granularity-aware evaluation for dialogue topic segmentation. arXiv:2512.17083.

Mostafa Dehghani, Yi Tay, Alexey A. Gritsenko, Zhe Zhao, Neil Houlsby, Fernando Diaz, Donald Metzler, and Oriol Vinyals. 2021. The benchmark lottery. *Preprint*, arXiv:2107.07002.

Jesse Dodge, Suchin Gururangan, Dallas Card, Roy Schwartz, and Noah A. Smith. 2019. Show your work: Improved reporting of experimental results. In *Proceedings of EMNLP-IJCNLP*, pages 2185–2194.

Chris Fournier. 2013. Evaluating text segmentation using boundary edit distance. In *Proceedings of ACL (Volume 1: Long Papers)*, pages 1702–1712.

Chris Fournier and Diana Inkpen. 2012. Segmentation similarity and agreement. In *Proceedings of NAACL-HLT*, pages 152–161.

Soichiro Fujita, Tsutomu Hirao, Hidetaka Kamigaito, Manabu Okumura, and Masaaki Nagata. 2020. SODA: Story oriented dense video captioning evaluation framework. In *Computer Vision – ECCV 2020*.

Michel Galley, Kathleen R. McKeown, Eric Fosler-Lussier, and Hongyan Jing. 2003. Discourse segmentation of multi-party conversation. In *Proceedings of ACL*, pages 562–569.

Maria Georgescul, Alexander Clark, and Susan Armstrong. 2006. An analysis of quantitative aspects in the evaluation of thematic segmentation algorithms. In *Proceedings of the SIGdial Workshop on Discourse and Dialogue*, pages 144–151.

Marti A. Hearst. 1997. TextTiling: Segmenting text into multi-paragraph subtopic passages. *Computational Linguistics*, 23(1):33–64.

Adam Janin, Don Baron, Jane Edwards, Dan Ellis, David Gelbart, Nelson Morgan, Barbara Peskin, Thilo Pfau, Elizabeth Shriberg, Andreas Stolcke, and Chuck Wooters. 2003. The ICSI meeting corpus. In *Proceedings of ICASSP*, pages I–364–I–367.

Junfeng Jiang, Chengzhang Dong, Sadao Kurohashi, and Akiko Aizawa. 2023. SuperDialseg: A large-scale dataset for supervised dialogue segmentation. In *Proceedings of EMNLP*.

Anna Kazantseva and Stan Szpakowicz. 2012. Topical segmentation: a study of human performance and a new measure of quality. In *Proceedings of NAACL-HLT*, pages 211–220.

Maurice G. Kendall. 1938. A new measure of rank correlation. *Biometrika*, 30(1/2):81–93.

Omri Koshorek, Adir Cohen, Noam Mor, Michael Rotman, and Jonathan Berant. 2018. Text segmentation as a supervised learning task. In *Proceedings of NAACL*, pages 469–473.

Ranjay Krishna, Kenji Hata, Frederic Ren, Li Fei-Fei, and Juan Carlos Niebles. 2017. Dense-captioning events in videos. In *Proceedings of ICCV*, pages 706–715.

Sylvain Lamprier, Tassadit Amghar, Bernard Levrat, and Frederic Saubion. 2007. On evaluation methodologies for text segmentation algorithms. In *Proceedings of ICTAI*, volume 2, pages 19–26.

Colin Lea, Michael D. Flynn, René Vidal, Austin Reiter, and Gregory D. Hager. 2017. Temporal convolutional networks for action segmentation and detection. In *Proceedings of CVPR*, pages 1003–1012.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*.

Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. 2024. The AI scientist: Towards fully automated open-ended scientific discovery. *Preprint*, arXiv:2408.06292.

Tengchao Lv, Lei Cui, Momcilo Vasilijevic, and Furu Wei. 2021. VT-SSum: A benchmark dataset for video transcript segmentation and summarization. arXiv:2106.05606.

Nitika Mathur, Timothy Baldwin, and Trevor Cohn. 2020. Tangled up in BLEU: Reevaluating the evaluation of automatic machine translation evaluation metrics. In *Proceedings of ACL*, pages 4984–4997.

Benjamin Minixhofer, Jonas Pfeiffer, and Ivan Vulić. 2023. Where's the point? self-supervised multilingual punctuation-agnostic sentence segmentation. In *Proceedings of ACL*, pages 7215–7235.

John Niekrasz and Johanna D. Moore. 2010. Unbiased discourse segmentation evaluation. In *Proceedings of the IEEE Spoken Language Technology Workshop*, pages 43–48.

Oriol Nieto, Morwaread M. Farbood, Tristan Jehan, and Juan Pablo Bello. 2014. Perceptual analysis of the F-measure for evaluating section boundaries in music. In *Proceedings of ISMIR*, pages 265–270.

Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. 2018. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606.

Mayu Otani, Yuta Nakashima, Esa Rahtu, and Janne Heikkilä. 2020. Uncovering hidden challenges in query-based video moment retrieval. In *Proceedings of the British Machine Vision Conference (BMVC)*.

Lev Pevzner and Marti A. Hearst. 2002. A critique and improvement of an evaluation metric for text segmentation. *Computational Linguistics*, 28(1):19–36.

Qwen Team. 2025. Qwen3 technical report. arXiv:2505.09388.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *Proceedings of the 40th International Conference on Machine Learning*, pages 28492–28518.

Colin Raffel, Brian McFee, Eric J. Humphrey, Justin Salamon, Oriol Nieto, Dawen Liang, and Daniel P. W. Ellis. 2014. mir_eval: A transparent implementation of common MIR metrics. In *Proceedings of ISMIR*, pages 367–372.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, pages 3982–3992.

Fabian Retkowski and Alexander Waibel. 2024. From text segmentation to smart chaptering: A novel benchmark for structuring video transcriptions. In *Proceedings of EACL (Volume 1: Long Papers)*, pages 406–419.

Fabian Retkowski, Maike Züfle, Thai Binh Nguyen, Jan Niehues, and Alexander Waibel. 2026. Beyond transcripts: A renewed perspective on audio chaptering. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8765–8787.

M. Saquib Sarfraz, Naila Murray, Vivek Sharma, Ali Diba, Luc Van Gool, and Rainer Stiefelhagen. 2021. Temporally-weighted hierarchical clustering for unsupervised action segmentation. In *Proceedings of CVPR*, pages 11220–11229.

Martin Scaiano and Diana Inkpen. 2012. Getting more from segmentation evaluation. In *Proceedings of NAACL-HLT*, pages 362–366.

Mike Zheng Shou, Stan Weixian Lei, Weiyao Wang, Deepti Ghadiyaram, and Matt Feiszli. 2021. Generic event boundary detection: A benchmark for event segmentation. In *Proceedings of ICCV*, pages 8055–8064.

Darshan Singh S, Anchit Gupta, C. V. Jawahar, and Makarand Tapaswi. 2023. Unsupervised audio-visual lecture segmentation. In *Proceedings of WACV*, pages 5221–5230.

Emiel van Miltenburg, Chris van der Lee, and Emiel Krahmer. 2021. Preregistering NLP research. In *Proceedings of NAACL-HLT*, pages 613–623.

Ramakrishna Vedantam, C. Lawrence Zitnick, and Devi Parikh. 2015. CIDEr: Consensus-based image description evaluation. In *Proceedings of CVPR*, pages 4566–4575.

Lucas Ventura, Antoine Yang, Cordelia Schmid, and Gül Varol. 2025. Chapter-Llama: Efficient chaptering in hour-long videos with LLMs. In *Proceedings of CVPR*, pages 18947–18958.

Teng Wang, Ruimao Zhang, Zhichao Lu, Feng Zheng, Ran Cheng, and Ping Luo. 2021. End-to-end dense video captioning with parallel decoding. In *Proceedings of ICCV*.

Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. 2020. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in NeurIPS*, volume 33, pages 5776–5788.

Huiyuan Xie, Zhenghao Liu, Chenyan Xiong, Zhiyuan Liu, and Ann Copestake. 2021. TIAGE: A benchmark for topic-shift aware dialog modeling. In *Findings of EMNLP*.

Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan, Kai Dang, Bin Zhang, Xiong Wang, Yunfei Chu, and Junyang Lin. 2025. Qwen2.5-Omni technical report. *Preprint*, arXiv:2503.20215.

Yi Xu, Hai Zhao, and Zhuosheng Zhang. 2021. Topic-aware multi-turn dialogue modeling. In *Proceedings of AAAI*.

Antoine Yang, Arsha Nagrani, Ivan Laptev, Josef Sivic, and Cordelia Schmid. 2023a. VidChapters-7M: Video chapters at scale. In *Advances in NeurIPS (Datasets and Benchmarks Track)*, volume 36, pages 49428–49444.

Antoine Yang, Arsha Nagrani, Paul Hongsuck Seo, Antoine Miech, Jordi Pont-Tuset, Ivan Laptev, Josef Sivic, and Cordelia Schmid. 2023b. Vid2Seq: Large-scale pretraining of a visual language model for dense video captioning. In *Proceedings of CVPR*.

A Model Details

The sentence encoder is all-MiniLM-L6-v2 with mean pooling. The document encoder has 12 RoPE transformer layers (384 dimensions, 8 heads, dropout 0.1) with a causal mask (online mode). Training used weighted cross-entropy (weights 1 and 2), AdamW at 2.5×10^{-5} with warmup and a cosine schedule, batches of about 115,000 tokens and 15 epochs; the best epoch was chosen on validation F1. On YTSeg test, the five seeds reached exact-match F1 44.26 to 44.81 and Pk 29.42 to 29.95; the offline variant of seed 0 reached F1 47.05. Retkowski and Waibel (2024) report 43.37 for their YTSeg-only offline model (their Table 3). We found no official implementation, so this is a comparable baseline, not a strict reproduction. Training took about 25 minutes per epoch on one 16 GB GPU.

Artifacts, licenses and compute. All data were used for non-commercial research, and no modified data are redistributed. YTSeg (Retkowski and Waibel, 2024) is under CC BY-NC-SA 4.0 and VT-SSum (Lv et al., 2021) under CC BY-NC-ND 4.0; the AMI (Carletta et al., 2006) and ICSI (Janin et al., 2003) corpora and annotations are under CC BY 4.0; SuperDialseg (Jiang et al., 2023) is listed as Apache-2.0 and TIAGE (Xie et al., 2021) as MIT. VidChapters-7M (Yang et al., 2023a) is released under MIT, with use subject to YouTube's terms and derivatives limited to non-commercial academic research (its datasheet). AVLectures (Singh S et al., 2023) is marked CC BY-NC 4.0 in its repository (whose license file is CC0-1.0), and its source, MIT OpenCourseWare, is under CC BY-NC-SA 4.0. The creators of DialSeg711 (Xu et al., 2021) state no license; it joins dialogues from MultiWOZ (MIT) and the Stanford multi-domain dialogue set, and we used it for evaluation only. The VidChapters transcripts were produced by its authors with Whisper (Radford et al., 2023). The video sets come from public YouTube or lecture videos whose rights stay with their owners; we used only released transcripts and annotations and did not screen them for personal or offensive content beyond what their creators did. Models: all-MiniLM-L6-v2, Qwen3-4B-Instruct-2507 and Qwen3-14B (Apache-2.0) and RoBERTa-base (Liu et al., 2019) (MIT); the VidChapters and Chapter-Llama evaluation code is under MIT. MiniSeg has about 44M parameters (22.7M in the sentence encoder) and the cross-encoder about 125M. All runs used one 16 GB consumer GPU; training took about 31 GPU hours for the five YTSeg seeds and about 3 for the six dialogue runs, and with inference and language-model runs the total was roughly 45 GPU hours.

B Preregistration Details

Approval and timing. The times below come from file modification times and the project's conversation logs (local time); none is externally timestamped. An AI agent wrote all four preregistrations. Test 1: an author read the draft and asked for it to be frozen unchanged; it was frozen at 18:27 on 2026-11-05, the scoring code was re-pinned after the deviation below at 11:12 on 2026-11-06, and the verdicts were written at 11:41. Test 2: at 19:13 on 2026-11-06 an author asked the agent to write a preregistration on VT-SSum train and run it; it was frozen at 19:16, and the verdicts were written at 19:33. Test 3: at 19:50 on 2026-11-06, after Test 2's results, an author asked the agent for a test on half B; the preregistration was frozen at 13:16 on 2026-11-07 and the verdicts were written at 13:33. In Tests 1 to 3 an author read the preregistration before scoring; the logs record instructions and file times, not reading, so for Tests 2 and 3 this rests on the author's account. Test 4 (dialogue): frozen at 05:16 on 2026-11-08, approved by the agent at 05:17 under authority an author had delegated, scored from 05:17 to 05:22; the authors read it after scoring.

Deviation in Test 1. The first scoring run stopped with an out-of-memory error before any result was written. The fix read each output array once instead of once per document; tables from the original and fixed code were identical on open data. Only that file's hash in the freeze list changed, and the authors approved the rerun. The 569 VidChapters videos with one sentence dropped out under the frozen rules.

Document counts. VidChapters-7M test has 8,200 videos; 7,997 have released transcripts, and 7,926 remain after removing videos that also appear in YTSeg, on which MiniSeg was trained. The frozen rules scored 7,357 of them (at least two sentences and one internal boundary); the duration analysis (Appendix E) needs at least three sentences and uses 7,267. Published evaluations (Appendix G) score all 8,200 from the released annotations. AVLectures has 350 segmented lectures; 348 were scored, and Table 14 uses all 350 because it rescores the authors' released outputs. VT-SSum train halves A and B had 3,693 and 3,683 files after the exclusions below; 11 in each had fewer than two sentences or no internal boundary, leaving 3,682 and 3,672 scored. YTSeg test has 1,448 videos, 1,447 with an internal boundary; the long-video analyses use the 1,403 whose chapter times map one to one onto sentence boundaries.

Exposure before the freeze. Exposure of sealed data before each freeze was limited to file structure and counts. Exposure to results was not: Test 3 was designed after Test 2 was scored, from an exploratory run on half A. Before test 2, a process outside the project (probably an editor indexer) opened 316 VT-SSum train files; they were excluded from both halves (143 from half A, 173 from half B).

The earlier VT-SSum test. Before the multi-dataset study, a separate preregistered test on VT-SSum test (958 videos) found MiniSeg minus even inconclusive at collar 1 (0.056, intervals 0.044 to 0.067, threshold 0.05), MiniSeg minus random supported (0.100), and the claim that placement matters more than count falsified: with its own YTSeg-fit threshold, the model predicted 2,429 of 32,600 boundaries. Its results motivated this study and are counted as open data.

C Further Results

Mean segment	YTSeg	VT-SSum
up to 6	$-.036^{*}$	$-.146^{*}$
6 to 10	.000	$-.099^{*}$
10 to 16	$+.113^{*}$	$+.012$
16 to 25	$+.267^{*}$	$+.056^{*}$
over 25	$+.243^{*}$	$+.119^{*}$

Table 6: Open data: MiniSeg minus even spacing, F1 at collar 3, by mean segment length in sentences (seed 0). *: 95% interval excludes 0.

Tolerances in words. To remove the sentence splitter from both the window and L, we rescored the oracle-count outputs with tolerances of 10, 25 and 50 words, grouped documents by mean segment length in words and placed even spacing by word position (post hoc; boundaries stay at sentence starts). The pattern held on every set. With top-K MiniSeg the contrast turned negative at $r \approx 0.7$ on VT-SSum (test: -0.037 at 25 words for segments up to 100 words) and at $r \approx 0.9$ on VidChapters. With peak suppression ($g = 3$) it stayed positive at $r \approx 0.7$ and turned negative only at $r \geq 1.3$, by -0.008 to -0.034, but the lead still shrank steadily with r (VT-SSum test, shortest group: $+0.135$, $+0.062$, -0.010 at $r = 0.29$, 0.71, 1.41).

Snapping. VidChapters chapter times are snapped to the nearest sentence start. The median shift was 1.1 seconds (median sentence 3.4 seconds) and 72% moved by at most 3 seconds, but 10% moved by over 23 seconds and 7.3% of boundaries merged. Poorly split videos concentrate in the shortest sentence group (median shift 11% of a chapter there, against 0.4% to 2% elsewhere; 18 words per sentence against 13.5 above 25). Keeping only videos whose every boundary moved by at most 10% of a chapter without merging (275 of 1,261 in that group), collar 3 still put MiniSeg 0.075 below even spacing (-0.103 to -0.047; with peak suppression -0.036), and WD favored it. YTSeg and VT-SSum labels are sentence-native and need no snapping.

Bootstrap units. Resampling VideoLectures.NET events (136 to 475 clusters) or speakers instead of videos left every VT-SSum interval almost unchanged (test, shortest group, collar 3: -0.167 to -0.135 by video, -0.167 to -0.133 by event).

Does WD reward bunching? On the open test splits we simulated two predictors with the same content: half of the reference boundaries exact and the rest either at random positions or next to the kept boundaries. Chance-corrected WD scored the bunched predictor lower in every length group but the longest VT-SSum one, most in the shortest groups (0.289 against 0.346 on VT-SSum, 0.262 against 0.395 on YTSeg). For MiniSeg itself, suppressing adjacent picks raised WD on whole sets by 0.012 to 0.033. WD thus penalizes bunching, so MiniSeg's WD lead is not produced by it.

Matching rule. Nearest-first matching is not always a maximum one-to-one matching: with references at 2 and 5, predictions at 4 and 7 and collar 3, it finds one match where two exist. Post hoc, we rescored the oracle-count outputs with maximum matching (predictions in order, each taking the eligible reference whose window ends first; checked against exact bipartite matching). No sign changed in any set, length group or metric, for MiniSeg top-K or with peak suppression. Maximum matching helps MiniSeg more because its bunched picks compete for the same reference: in the shortest group it added about 4% to MiniSeg's collar-3 matches and 1% to even spacing's, shrinking the reversal on VT-SSum test from -0.152 to -0.139 (interval -0.154 to -0.122) and on VidChapters from -0.066 to -0.063. One interval changed status, toward a stronger reversal (YTSeg, peak suppression, shortest group: -0.034, -0.050 to -0.014). Local relative F1 was identical, because its windows never overlap.

Metric	Half range	Tau	Below even
WindowDiff (selected)	0.033	0.858	0
local 0.2	0.046	0.920	0
Pk	0.058	0.858	0
Collar 0	0.065	0.893	0
Collar 1	0.174	0.769	1
Collar 3	0.258	0.570	3
B, $n_t = 2^*$	0.056	0.840	0
B, $n_t = 5^*$	0.088	0.662	2

Table 7: Metric comparison on the open test splits (chance corrected, seed 0). Half range: range of the half-correct predictor's score over the ten dataset and length groups. Tau: mean pairwise Kendall tau (Kendall, 1938) of five methods' order across groups. Below even: groups, of 10, where MiniSeg scores below even. *Added after the selection.

Mean segment	Videos	C1	C3	WD
up to 6	1,261	+.036	−.066	+.256
6 to 10	1,123	+.201	−.027	+.270
10 to 16	1,324	+.259	+.084	+.251
16 to 25	1,123	+.266	+.164	+.218
over 25	2,526	+.194	+.168	+.179

Table 8: VidChapters (sealed): MiniSeg minus even spacing by mean segment length, medians over five seeds. Every interval excludes 0.

Meetings. AMI levels 1 and 2 and ICSI levels 1 to 3 and Galley's reference favored MiniSeg over even spacing on every metric in point estimates (collar 3: $+0.124$ to $+0.165$; WD: $+0.139$ to $+0.345$). AMI's collar-3 advantage was larger at the coarse level, with overlapping intervals; ICSI showed no such rise. Several ICSI and Galley intervals include 0.

Set	Count rule	$/K$	≤ 6	> 25	WD
VT-SSum B	threshold	1.17	−.078	+.065	+.134
VT-SSum B	length	0.94	−.047	−.016	+.154
VT-SSum test	threshold	1.14	−.067	+.050	+.142
VidChapters	threshold	0.62	−.002	+.151	+.195
YTSeg test	threshold	1.00	−.014	+.230	+.296

Table 9: Calibrated counts (post hoc, seed 0): MiniSeg minus even, collar 3 by mean segment length and WD over all videos. Threshold: MiniSeg margin threshold at which the dev split's total count equals its reference total (VT-SSum dev for VT-SSum, YTSeg validation otherwise). Length: $K = \mathrm{round}(n/L) - 1$ with L the dev mean segment length. $/K$: median count over reference count.

Test 3 by length. In test 3 (Figure 2), C99 showed the MiniSeg pattern at a smaller scale: collar 3 ran from -0.033 for videos up to six sentences to $+0.072$ above 25, while WD stayed between $+0.057$ and $+0.065$. TextTiling was below even spacing at collar 3 in every group, so it is weak here, but its gap still shrank with length by more than its WD gap did (T6). Seed variation was small next to bootstrap variation: on VidChapters, point estimates of the WD contrast ranged from 0.224 to 0.229 across seeds.

Other fixed windows (post hoc except V5, V6). The reversal is a property of fixed windows, not of F1. Boundary similarity B with its default near-miss window of two sentences, the metric YTSeg reports, did not reverse, but widening the window to five sentences brought the reversal back, as test 2 confirmed (V5, V6); in the shortest group even B_2 lost almost all of MiniSeg's advantage ($+0.006$). Post hoc, we rescored VidChapters in seconds against even spacing in time: the usual 3 and 5 second tolerances, strict next to its chapters (median 110 s), favored MiniSeg in every group, but at 30 seconds MiniSeg fell below even spacing for chapters up to two minutes (Table 10). What matters is the width of the tolerance relative to the segments, in sentences or in seconds.

D Language Model Details

Model: Qwen3-4B-Instruct-2507 in bf16 with greedy decoding and at most 120 new tokens, on one 16 GB GPU (median 21 s per VT-SSum video

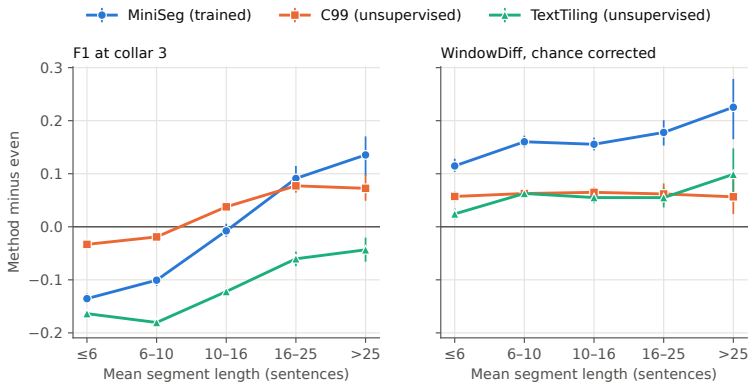


Figure 2: Test 3 (unread VT-SSum train videos, reference K): method minus even spacing by mean segment length. Medians over five runs; bars span the 95% intervals across runs. Collar 3 is lowest in short segments and highest in long ones for all three methods, while WindowDiff stays positive in every group (nearly flat for C99).

Mean chapter	Videos	3 s	5 s	10 s	30 s
up to 60 s	1,769	+.209	+.179	+.071	−.104
60–120 s	2,203	+.243	+.240	+.200	−.022
120–240 s	1,689	+.174	+.180	+.178	+.078
240–480 s	975	+.122	+.132	+.143	+.109
over 480 s	721	+.095	+.107	+.119	+.118

Table 10: VidChapters (sealed outputs, rescored post hoc): MiniSeg minus even spacing in time, F1 by tolerance in seconds and mean chapter duration, medians over five seeds. Every interval excludes 0.

Set ($\leq$ 6 sentences)	Top-K	g=1	g=2	g=3	g=4	g=5
VT-SSum test	−.152	−.065	−.031	−.031	−.053	−.079
VT-SSum half A*	−.137	−.056	−.029	−.033	−.055	−.075
VT-SSum half B*	−.135	−.057	−.025	−.028	−.051	−.071
VidChapters*	−.066	−.026	−.022	−.027	−.036	−.046
YTSeg test	−.036	−.008	−.008	−.020	−.030	−.042
VT-SSum test, 6–10	−.102	−.037	−.002	+.017	+.028	+.022
VT-SSum test, all	−.080	−.014	+.017	+.029	+.028	+.017

Table 11: Peak suppression, a fixed-radius non-maximum suppression (post hoc): MiniSeg minus even spacing, collar-3 F1 at the reference count, when picks within g sentences of an earlier pick are skipped; medians over five seeds. *: sealed sets already scored. In the shortest group every interval excludes 0 except YTSeg (131 videos) at $g \leq 3$. WD point estimates favored MiniSeg in every group and radius.

and 6 s per YTSeg video). The 14B rerun used Qwen3-14B in 4-bit NF4 with bf16 compute and thinking mode off, otherwise unchanged (median 36 s and 12 s; every window parsable). Each sentence was cut to its first 60 words. The prompt:

> Below is an excerpt from a video transcript, with numbered sentences. List the numbers of the sentences in the excerpt where a new topic or chapter begins, that is, where the speaker clearly moves on to a different subject. Most excerpts contain few such sentences, and some contain none. [excerpt]
>
> Answer with one JSON list of integers (possibly empty) and nothing else.

Prompts tried first, on the whole transcript, asked for exactly K boundaries, for chapter titles with start sentences, or for a free number of boundaries; on long inputs the model listed consecutive sentence numbers or repeated the transcript. In the windowed prompt, lists cut off by the token limit were parsed as far as they went; in the final run every window gave a parsable list. Ties were broken by seeded noise. On YTSeg, MiniSeg (trained on YTSeg) beat the language model on every metric; on VT-SSum the two were close (WD over all videos: +0.152 and +0.133). The 14B model beat MiniSeg on VT-SSum (+0.208) but not on YTSeg (+0.310 against +0.372).

Results. We prompted Qwen3-4B-Instruct-2507 (Qwen Team, 2025), without training, to list the sentences where a new topic begins in windows of 60 numbered sentences (stride 20); a sentence's score is the share of covering windows that listed it, and the top K are kept. Asking for exactly K boundaries over a whole transcript failed on long inputs, so we chose this prompt on about a dozen videos by whether outputs were usable, not by scores (below). On 150 videos each from the opened VT-SSum and YTSeg test splits, 30 per length group, the language model showed the same reversal as MiniSeg (Table 5). With its own count (sentences listed by at least half their windows), it placed about half of K in the shortest group and two to five times K in the longest, and collar 3 still put it below even spacing in the shortest group on both sets. Post hoc, under a plan written before generation, we reran the same sample and prompt

with Qwen3-14B (4-bit). The reversal held in the shortest group on both sets but was about half as large; the larger model beat even spacing by more within one sentence ($+0.07$ and $+0.12$, against -0.05 and $+0.06$), which by Eq. 1 should shrink the reversal. With its own count, its collar-3 deficit there was no longer significant.

E Duration Analysis with Our Metrics

Metric	Predictor, set	< 10	≥ 60	Change
F1, 6 s	Constant, YTSeg	.704	.270	-62%
	Constant, VidCh.	.772	.366	-53%
	MiniSeg, YTSeg	.613	.304	-50%
	MiniSeg, VidCh.	.493	.233	-53%
Local rel. F1	Constant, YTSeg	.702	.732	$+4\%$
	Constant, VidCh.	.717	.696	-3%
	MiniSeg, YTSeg	.486	.243	-50%
	MiniSeg, VidCh.	.315	.143	-55%
WD	Constant, YTSeg	.687	.719	$+5\%$
	Constant, VidCh.	.737	.729	-1%
	MiniSeg, YTSeg	.384	.237	-38%
	MiniSeg, VidCh.	.258	.162	-37%

Table 12: Sentence-level scores by video duration in minutes at the reference count (post hoc, seed 0). F1 at 6 s uses nearest-first matching within 6 seconds (raw); local relative F1 and WD are chance corrected. Without chance correction MiniSeg's declines are smaller (local: -34% and -31%; $1-$WindowDiff: -13% and -8%). Constant: relative error $s = 0.15$, snapped to the nearest sentence.

YTSeg test (1,403 videos), VidChapters (7,267) and AVLectures (348) were grouped by duration. The median segment lasted 56 and 59 seconds in videos under 10 minutes and 425 and 565 seconds in videos over an hour (YTSeg, VidChapters). Chance correction against random placement did not remove the 6-second trend for the simulated predictor (-62% and -57%). On YTSeg and VidChapters, F1 at a 6-second tolerance fell by 53% to 62% for the simulated predictor and 50% to 53% for MiniSeg, while chance-corrected local relative F1 and WD stayed within 5% for the simulated predictor and fell by 50% to 55% and by 37% to 38% for MiniSeg. On AVLectures, with segments of about nine minutes, rankings reversed the other way: in lectures over an hour, the simulated predictor beat MiniSeg on local relative F1 (0.80 against 0.43) but scored below it at 6 seconds (0.08 against 0.18).

F Coverage Model and Human Annotations

Coverage model. Bunched picks may explain part of the shortfall in the shortest segments: 30.7% of consecutive MiniSeg picks were adjacent on VT-SSum, against 14.7% of reference boundaries, but peak suppression did not close the gap: suppressing picks within three sentences of an earlier pick, a rule chosen after seeing these results, turned VT-SSum's overall collar-3 contrast positive ($+0.029$) but not the shortest group's (-0.030). For gamma-distributed segment lengths with mean L and shape $a = 1/\mathrm{CV}^2$, $\mathbb{E}[(G - w)_+] = L\,Q(a{+}1, aw/L) - w\,Q(a, aw/L)$, with Q the regularized upper incomplete gamma function; $\mathrm{CV} \to 0$ gives $\min(1, w/L)$ and $\mathrm{CV} = 1$ gives $1 - e^{-w/L}$. Equation 1 treats the even grid's phase as random relative to the references, but both document ends pin it. Simulating gamma segment lengths with maximum matching, the equation under-predicts even spacing when segments are regular and boundaries few: at $\mathrm{CV} = 0.3$ and $r = 0.5$, simulated F1 was 0.66 for $K = 3$, 0.54 for $K = 10$ and 0.50 for $K = 100$, against 0.50 from the equation; at $\mathrm{CV} \geq 0.6$ and $r \leq 1$ the gap stayed within 0.03 for every K. It also ignores two even boundaries competing for one reference, so it over-predicts by up to 0.06–0.09 when $w > L$. A simulation that reorders each document's own segment lengths at random predicted even-spacing F1 within 0.005, so the order of segment lengths hardly matters. Under the alignment view the error of the j-th even boundary is a discrete Brownian bridge with standard deviation about $\mathrm{CV} \cdot L\sqrt{K+1}/2$ at the middle boundary; it exceeded three sentences in 87% to 99% of documents. All of this is post hoc and uses seed-0 outputs; p is fitted at collar 1, so only collars 2 and 3 test the MiniSeg side.

Human annotations. ICSI's NXT topic segmentation holds the files of several coders; we used Galley's segmentation against the coder whose files the main analysis scored (levels 1 and 2, 25 meetings) and pairs of coders (10 meetings), both directions, dialogue acts as units. Mean reference segments were 125 to 287 dialogue acts, so large c/L comes from widening the collar, not from short segments. Against even spacing at the reference count instead of the human's count, the advantage turned negative at $c/L = 1$ in three of four pair sets, significantly in two (e.g. -0.089, 95% interval -0.165 to -0.014), partly because annotators chose different

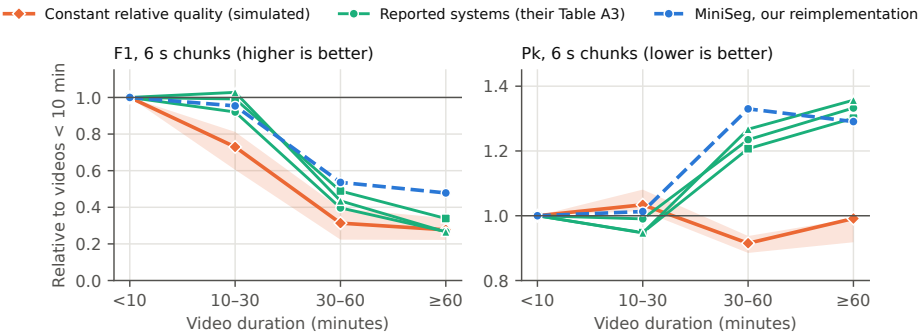


Figure 3: Scores by video duration under the `chunkseg` scorer (6-second chunks), each divided by the same method's score on videos under 10 minutes. Orange: simulated predictor of constant relative quality ($s = 0.10$; band: s from 0.05 to 0.25). Green: the three systems that Retkowski et al. (2026) report in all four duration bins, from their Table A3. Blue: our MiniSeg. F1 falls as much for the simulated predictor as for the published systems; Pk stays flat for the simulated predictor but rises by about a third for real systems.

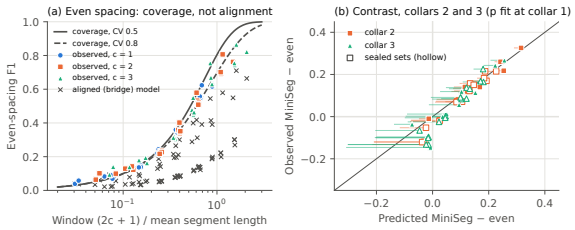


Figure 4: (a) Even-spacing F1 against window width relative to mean segment length: observed values, the coverage model (Eq. 1) for two coefficients of variation, and the alignment model. (b) Predicted against observed MiniSeg minus even at collars 2 and 3 (collar 1 fits p and is omitted); bars run from a model with no gain beyond collar 1 to the mixture model; hollow: sealed sets.

counts. Galley's segmentation is a consensus of several judges and is related to the NXT coding, so pairs may not be fully independent. Intervals: cluster bootstrap over meetings.

Mean segment	YTSeg	VidChapters	VT-SSum
up to 6	+.126	+.036	−.047
6 to 10	+.312	+.201	+.085
10 to 16	+.334	+.259	+.146
16 to 25	+.372	+.266	+.152
over 25	+.260	+.194	+.143

Table 13: MiniSeg minus even spacing at collar 1 by mean segment length (post hoc). YTSeg: in domain; VidChapters: same label source, new videos (median over seeds); VT-SSum: slide-derived labels.

What the ratio does not explain. At equal segment length, MiniSeg's collar-1 advantage on YTSeg exceeded VT-SSum's by 0.11 to 0.23, with VidChapters about halfway (Table 13). About half of the gap between datasets therefore comes from new videos or training on YTSeg, and half coincides with slide-derived labels and lecture content (post hoc, observational).

G Published Evaluations and Tables

Even spacing under published evaluations. All analyses here are post hoc. VidChapters-7M test (8,200 videos; already scored in test 1): we ran the original evaluation files of Yang et al. (2023a) and Ventura et al. (2025) unmodified and our reimplementation, which agreed. The ground truth was rebuilt from the released annotations without the authors' tokenizer filter; restricting to the 8,191 videos Chapter-Llama scores changed results by at most 0.05. Chapter-Llama's proprietary-model rows use a 10% test subset that is not public. YTSeg test: random placement with the reference count reproduced the published random baseline of Retkowski et al. (2026) on B (7.8 against 7.90) and Pk (47.8 against 48.43); F1 depends on how chunk times are derived (7.1 or 11.1 against 8.57). AVLectures: our port of the authors' notebook reproduced their stored outputs to three decimals, including their equal-split and full-method rows. The label-free count is a log-log fit of segment count on duration over training videos; AVLectures has no training split, so its count used the mean segment duration of the other 14 courses. Intervals: bootstrap over channels or courses (1,000 draws).

Under the Table-4 scorer at tIoU 0.3, 0.5, 0.7 and 0.9, even spacing with the reference count scored 88.6, 55.5, 24.5 and 5.4 F1 (with the count from duration 79.2, 43.2, 16.1 and 3.0; random placement

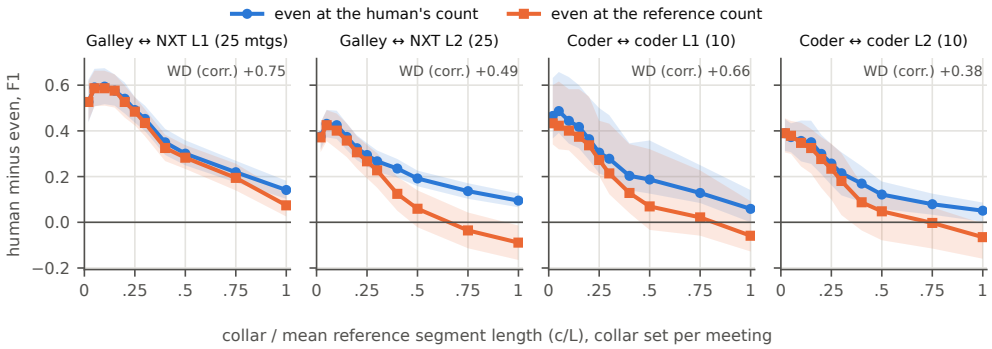


Figure 5: ICSI, human topic segmentations scored against each other (post hoc): collar F1 of the human hypothesis minus even spacing, by collar relative to mean reference segment length. Bands: 95% cluster-bootstrap intervals over meetings. WD (corr.): chance-corrected WindowDiff, human minus even at the human's count.

Mean segment	BS@30	BS@120	BS@300	F1
under 6 min (73)	+30.9	+6.8	−1.5	+6.5
6–9 min (94)	+33.0	+12.9	−5.0	+3.8
9–12 min (61)	+21.9	+15.6	−1.5	+1.8
12 min or more (122)	+20.1	+19.1	+1.8	+0.5
All (350)	+26.1	+14.3	−1.3	+2.9

Table 14: AVLectures: the method of Singh S et al. (2023), rescored from its released outputs, minus even spacing at the reference count, by boundary window and mean segment length (lectures in parentheses). The released outputs cover all 350 segmented lectures, including the two our other analyses drop (Section 3). F1: segment-overlap F1.

with K 69.5, 36.7, 14.1 and 2.8). The published tables give only 0.5 and 0.7. At 0.7, even spacing with K was below four fine-tuned Vid2Seq variants (24.7 to 27.4), tied PDVC (24.5) and was above the visual-only variants. Chapter-Llama's appendix table (its Table A.13) gives Chapter-Llama 64.8 and 48.8 at 0.5 and 0.7, above even spacing at both, and Vid2Seq 48.6 and 29.6, which differ from the VidChapters paper's numbers for the same model; against those, label-free even spacing (43.2) does not tie Vid2Seq. **Count or spreading?** Even spacing with K differs from even spacing with the duration count only in the count, and from random placement with K only in placement. At tIoU 0.3, 0.5, 0.7 and 0.9 the count was worth 9.5, 12.3, 8.4 and 2.4 points and spreading 19.1, 18.8, 10.4 and 2.7 (channel-bootstrap intervals within ± 1); label-free even spacing beat random placement with the true count by 9.6, 6.5 and 2.0 points up to 0.7. Spreading paid only where chapters are regular: in videos whose chapter durations had CV below 0.5 (3,453 of 8,200) it was worth 30.5 points at 0.5, at CV 0.5 to 0.8 16.1, and above 0.8 even spacing fell below random placement (-0.9 and -6.2), as the CV term of Equation 1 implies. These comparisons cover localization only: even spacing writes no titles, so title-aware metrics such as SODA$_c$ (Fujita et al., 2020) or CIDEr (Vedantam et al., 2015) do not apply.

On VidChapters at 5 seconds, even spacing with the reference count (20.2) was close to random placement with the same count (19.4), and much of its score came from a boundary at time zero. Under temporal IoU, even spacing was nearly flat across video durations (Chapter-Llama F1 25.9 to 23.6 from short to long videos), so Vid2Seq's decline from 33.4 to 16.7 is a real loss of that system; mean tIoU has a high floor (random placement with the reference count scored 52). On YTSeg videos under 30 minutes, Qwen2.5-Omni (Xu et al., 2025) in its default setting scored below even spacing (F1 3.4 against 8.8; Table 7 of Retkowski et al. (2026)), and Pk at 6-second chunks did not separate zero-shot WtP (Minixhofer et al., 2023) (47.4) from even spacing (47.6) although WtP's F1 was four times higher. On AVLectures without the reference count, the authors' method with an estimated count scored F1 42.6 and 42.7 (their supplementary Table 7), against 48.4 (43.1 to 54.3) for label-free even spacing, which was equal on MoF and lower on NMI and at 30 seconds. Per course (15 courses), the method led even spacing with K in all 15 at ± 30 and ± 120 seconds and in 7 at ± 300 seconds; against the label-free count, in 15, 15 and 9. Under the Chapter-Llama scorer, fine-tuned Vid2Seq scored 19.0 F1 on 15 to 30 minute videos against 24.2 for even spacing with K and 15.3 without labels.

Published tables. We searched for papers that report the same systems under several boundary tolerances or metrics, across video chaptering and dense captioning (seconds collars, temporal IoU thresholds), generic event boundary detection (relative distance 0.05 to 0.5), action segmentation (F1@10, 25, 50; Lea et al., 2017) and text, dialogue and meeting segmentation (F1, Pk, WD, B). We transcribed 76 tables from 30 papers (4,098 values, 28 papers from their LaTeX sources), each checked against its source. A pair of systems flipped if its order differed between two settings; ties at the printed precision did not count. Between the tightest and widest tolerance of 45 tables, 14.0% of pairs flipped (9.7% when both gaps were at least 0.5 points) and the top system changed in 9 tables; F1 and WD disagreed on 22.1% of pairs, F1 and Pk on 16.2%, Pk and WD on 8.7%. Seconds collars in chaptering, at 2% to 8% of a chapter, flipped 2.8% of pairs and never changed the top system. Across all comparisons the flip share rose with tolerance relative to segment length (Spearman $\rho = 0.32$), but more with the distance between the two settings ($\rho = 0.66$). Examples of claims that held at one reported setting and failed at another: in Table 2 of Yang et al. (2023a), zero-shot shot detection beat the best fine-tuned model on precision at 5 seconds (33.2 against 30.3) but not at tIoU 0.5 (18.0 against 43.1); on TAPOS, three event-boundary detectors claimed the state of the art, which held at tight relative distance and not at 0.5; in Coen (2025), evenly spaced boundaries with the gold count ranked third of four on strict F1 (0.276) and first on $\pm$1-utterance F1 (0.633 against 0.628). These tables were selected for reporting several settings, so the rates are not population estimates, and event-boundary papers share baseline rows.

H Prospective Test on Dialogue

Design. The unit is the utterance. SuperDialseg (6,863 train, 1,305 dev and 1,310 test dialogues; 1,269 test dialogues with an internal boundary scored) is the primary set, and the models were trained on its train split only. TIAGE (97 of 100 test dialogues scored) is out of domain for the models. DialSeg711 (702 test dialogues) has only 7 development dialogues, so no p was fitted and it enters only the even-spacing and WD claims. MiniSeg used our code with utterances as units (dev exact F1 83.4 to 83.6); the RoBERTa-base cross-encoder reached 84.9 to 85.6. Every method placed the reference count. The share p was fixed per model seed, set and length group on development data. Predictions use each test group's observed L and coefficient of variation, which the frozen scorer computes from the reference segment lengths; no method output on test enters a prediction. Development data showed before the freeze that Equation 1 in closed form over-predicts even spacing in this regime, so an exact simulation from each dialogue's own segment lengths in random order became the primary even-side predictor, and the closed form was kept as a separate claim expected to fail. Thresholds came from the post hoc numbers in Section 8 (correlation 0.96 to 0.8, sign agreement 0.93 to 0.8, mean absolute error 0.017 to 0.03).

Procedure. The preregistration, code, frozen p, prepared and raw test data and all checkpoints (39 files) were pinned by SHA-256 before scoring. The test files had been hashed but not opened; the DialSeg711 README shows one example dialogue, and the two test dialogues starting with its first utterance were excluded. Five MiniSeg seeds were planned and cut to three before scoring for GPU time, and one seed was retrained from scratch after its first run was stopped by mistake; both are disclosed in the preregistration. Approval was given by the AI agent under the authors' delegated authority (Appendix B); the authors reviewed the preregistration after scoring and found no problem. Its claims are counted with those of Tests 1 to 3 (Table 2). The sealed run was scored once, with no deviation, and the freeze still matched afterwards.

Claim (model − even)	MiniSeg	Cross-enc.	Verdict
H1: C2, C3 F1 < 0, $L \leq 2.5$	$-.009, -.010$	$-.011, -.013$	S, S
H2: WD > 0, every group	$\geq +.51$	$\geq +.57$	S, S
H3: corr $> .8$, sign $\geq .8$	.83 / .74	.84 / .76	F, F
H4: signs where $\lvert\text{pred}\rvert > .02$	11/14 S, 0 F	10/12 S, 0 F	I, I
H5a: Eq. 1 MAE $< .03$	.077 (bias $+.076$)		F
H5b: simulation MAE $< .03$	.022 (.019–.033)		I

Table 15: Preregistered dialogue claims (medians over three runs; verdicts use the extreme 95% bounds across runs). S supported, F falsified, I inconclusive. Secondary claims: WD > 0 on TIAGE and DialSeg711 was inconclusive for both models; H3 with pooled p was falsified for both.

Where the predictions failed. Of the nine sign errors among cells predicted above 0.02 in size, three are SuperDialseg collar-2 cells where the flip was predicted too early (-0.034 and -0.050 predicted, $+0.006$ and $+0.018$ observed), and six are

in TIAGE's 5 to 8 group, where the development fit clipped p at 0 because the models scored below the chance mixture out of domain. Every one of the nine observed intervals includes 0. The reversal at $L \leq 2.5$ is small because p is about 0.96 there and even spacing reaches 0.995.

Label structure. SuperDialseg segments are built from changes of grounding document, and its boundaries fall almost only before user turns. Both trained models learn this regularity, so the development share p is high (0.84 to 0.88 pooled over groups), and dialogue does not test a weak in-domain model.

I Decision Replay

All analyses here are post hoc and reuse outputs that had already been scored; no unscored set was read. A decision picks the best candidate on a development set by one criterion; its cost is the held-out score of the collar-3 (C3) pick minus that of the WD pick, with a cluster bootstrap. Only the best and last checkpoints of each run were saved, so checkpoint choice was replayed between those two and over training-log epochs only.

Decision (dev → test)	Differ	ΔWD	ΔC3
MiniSeg vs even, K; VT-SSum dev → test	5/5	$-.144$	$+.080$
→ half A* / half B*	5/5	$-.146$ / $-.150$	$+.073$
Label-free menu; VT-SSum dev → test, A*, B*	5/5	$-.153$ to $-.156$	$+.060$
MiniSeg vs even, K; YTSeg val → YTSeg, VidCh.*, AVL.*	0/5	0	0
Short YTSeg val → short YTSeg test (expected)	2/5	$-.162$	$+.002$
Best vs last checkpoint (24 decisions)	5/24	$\geq -.002$	
Peak-suppression radius		$-.002$ to $-.012$	

Table 16: Cost of choosing by collar-3 F1 instead of WD (medians over seeds). Differ: seeds in which the two criteria chose differently. *: sealed sets already scored in the preregistered tests. Expected: averaged over development bootstrap draws.

By mean segment length, with the reference count the choice flipped in every VT-SSum group up to 16 sentences ($r \geq 0.57$) and in the YTSeg and VidChapters group up to 6 ($r \geq 1.5$), never above 16 ($r \leq 0.37$); the YTSeg group at 6 to 10 sentences ($r \approx 0.85$) did not flip, so the ratio is not the only factor. In the label-free menu, collar 3 also chose C99 or even spacing with a duration-based count at 16 to 25 VT-SSum sentences, whose WD was below random (-0.010 to -0.052), at a cost of 0.14 to 0.20 WD; above 25 sentences no choice changed in any menu. On short subsets, collar 3 and WD gave significant opposite verdicts in 20 of 25 runs for MiniSeg against even spacing, 15 of 25 against C99 and 5 of 5 for C99 against even spacing, and in 0 of 30 on long subsets. Both language models lost to even spacing under collar 3 on short videos and beat it under WD. When peak-suppressed MiniSeg was among the candidates, both criteria chose it and the cost was at most 0.006 WD: the large cost needs a spreading baseline among the candidates.

J Even-Adjusted F1

For $\hat{K}$ predicted and K reference boundaries, $F_1^{\text{adj}} = (F_1 - F_1^{\text{even}})/(F_1^{\max} - F_1^{\text{even}})$ with $F_1^{\max} = 2\min(K, \hat{K})/(K + \hat{K})$ and counts pooled over the corpus (post hoc). Within a group all methods share F_1^{even}, so the adjustment is an affine map of raw F1 and cannot change a ranking: Kendall tau, the number of groups with MiniSeg below even spacing and agreement with WD equal raw F1's, and the score of the half-correct predictor varied more across lengths (1.230 on open sets, against 0.248 for raw collar 3 and 0.034 for WD). Its use is in reading published numbers, because F_1^{even} follows from segment statistics: the best VidChapters Vid2Seq is $+0.16$ at 5 seconds but -0.22 at temporal IoU 0.5, where every published system is negative; AVLectures' method is $+0.39$ at ± 30 seconds and -0.09 at ± 300. MiniSeg and C99 were never below even spacing where the headroom $1 - F_1^{\text{even}}$ was at least 0.6 (0 of 94 cells) and were below it in 36 of 62 cells below that; r separated these cells as well (AUC 0.987 against 0.974). A scorer that reports F_1^{even}, the headroom and r beside fixed-tolerance F1 will be released with the code.

K Errors Found During the Research

The work was run largely by autonomous AI agents. It began as a study of when to rerun a causal segmenter on a growing transcript under a budget of calls, where each call reveals the model's output for the two most recent sentences only. Several errors in evaluation code would have changed conclusions had they not been found. (1) Three runs let a call reveal more than those two sentences. (2) One run used the output of the model run over the whole transcript at sentences where no call was made, which overstated a recall ratio (0.830 instead of 0.35). (3) One run compared a raw output score with 0.5 instead of comparing the two class scores, which halved the model's recall. (4) A baseline named "equal duration" actually spaced boundaries evenly by sentence index. (5) A schedule that used the labels covered 95% of reference boundaries within one sentence; we first read this as showing

that the call budget was no barrier, but few of those boundaries fell in the sentences its calls revealed. In the sealed run, an out-of-memory error was fixed before any result was written. We report these because evaluation code, not only models, can drive conclusions.